
The modeling and dynamic analysis of the Kresling Conical Origami

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Abstract

Multi-stable origami structures have attracted wide attention across various fields due to their unconventional mechanical properties, such as reconfigurability and tunable stiffness. However, current studies mainly focus on the static properties of multi-stable origami structures. To advance the state of the art, this research chooses the Kresling conical origami (KCO) as the platform. Different from the normal Kresling structure, the KCO possesses three stable states and a piecewise non-smooth constitutive curve. The whole KCO structure could be equivalently represented as a nonlinear spring-damper system, and its dynamic model is reconstructed using a piecewise system identification method. Based on the model, the numerical simulation could be performed. The bifurcation analysis with respect to external excitation amplitude reveals various response types of the KCO structure, along with the transition processes between them. Furthermore, we also analyze the basin of attraction to understand how different responses evolve with changes in initial conditions. Overall, this study proposes a systematic approach for investigating the dynamics of multi-stable origami structures and provides a theoretical foundation for the control of their dynamic behaviors.

The Keywords Origami, Nonlinear dynamics, non-smooth system, piecewise system identification

1 Introduction

Origami structures have demonstrated revolutionary potential in various engineering and scientific fields due to their lightweight, foldability, and shape adaptability, including materials science[1], biology[2], robotics[3], and architecture[4]. For example, biologists have used origami to explain how leaves fold and unfold[5]; aerospace engineers have employed origami structures to design various types of foldable solar panels[6]; roboticists have used the rich deformability of origami structures to design various prototypes, including flexible robotic arms[7] and snake-like robots[8]. In recent years, many studies have been carried out on the mechanical properties of origami structures. Researchers have found that the folding motions of origami can lead to many unconventional mechanical properties, including negative Poisson ratio[9], [10], self-locking[11], [12], multi-stability[13], and tunable stiffness[14].

Among these mechanical properties, multi-stability has attracted particular attention from researchers. Compared with monostable origami structures, multi-stable ones offer three advantages: (1) rapid switching between stable configurations[15]; (2) no external force is needed to maintain the structure in each stable state[16]; (3) different stable configurations usually possess distinct geometric and mechanical properties[17]. Using these advantages, the multi-stable origami has wide applications in robotics, metamaterial science, and deployable structure designs. For instance, Wang et al. designed a modular multi-stable deformable structure based on a rigid origami unit, which could achieve reconfigurability and self-locking properties[18]. Kaufmann *et al.* proposed a soft robotic arm based on a bistable Kresling structure, and it could achieve lower control requirements and higher motion accuracy[19]. In addition, certain deployable structures can also be constructed using bistable origami sheets, such as Miura patterns, to enhance load-bearing capacity[20]. However, despite the

rich outcomes on multi-stable origami structures, the majority of studies are still limited to quasi-static analysis—primarily focusing on the static properties in various stable states—whereas the exploration of dynamic behaviors remains at a preliminary level. Noting that in practice, no matter origami robot or metastructure would inevitably work in a dynamic environment. Hence, extending the research from quasi-static to dynamic is essential. For example, Wu et al. investigated the axial transient deployment process of Miura-ori tubes and found the presence of vertical displacement[21]; Yasuda et al. designed a Kresling-based metastructure that could form rarefaction solitary waves and achieve impact absorption[22]. The above issues cannot be studied using static methods; hence, it is necessary to develop corresponding dynamic approaches.

The dynamics of multi-stable origami structures mainly focus on two aspects: the modeling method and the analysis process. The traditional modeling methods are always based on the first principle, and they treat rigid and non-rigid origami structures differently. The spatial bar and hinge dynamic modeling approach applies to rigid non-foldable, partially rigid-foldable, and fully rigid-foldable origami structures[22], [23]. The modeling process is convenient, but it ignores the folding characteristics. Finite element methods are also applicable to various types of origami structures[24], offering high accuracy but low computational efficiency. The multibody dynamic modelling method assumes that the facet is a rigid body, while the crease is an elastic hinge[25]; hence, this method is suitable for rigid origami. Overall, different methods have different ranges of applicability. More importantly, determining the geometric and material parameters required for origami modeling remains a major obstacle. For the analysis process, the multi-stable origami structure always possesses rich dynamic responses. Under small-amplitude external excitation, multistable origami structures typically exhibit small-amplitude intra-well vibrations near one of the stable states[26], [27]. As the amplitude of the external excitation continues to increase, the origami structure will trigger large-amplitude inter-well oscillations between different stable states[28], [29], and even subharmonic, superharmonic, and chaotic responses may occur[26], [30]. The occurrence of these dynamic responses is closely related to the amplitude and frequency of the external excitation, making corresponding analysis essential.

Based on the challenges mentioned above, this research aims to systematically model the multi-stable origami structure and analyze its dynamic characteristics. Here, we choose the Kresling conical origami (KCO) as the research platform. Different from the conventional Kresling origami[31], the KCO structure could possess three stable states through careful geometric parameter design. **The analysis of the KCO structure is still in its infancy. However, for its derivative structure, the Kresling structure, existing research is generally based on the bar-hinge model. For example, Cai et al[32] established a bar-hinge model for the Kresling structure and analyzed its constitutive properties using the finite element method. In another research, Agarwal et al[29] established the bar-hinge model for the Kresling structure and conducted research on its dynamic characteristics based on this model. In this research, to ensure the simplicity of the model,** the KCO structure could be regarded as a nonlinear spring-damper system, and the parameter identification method can be used to reconstruct the dynamic model. For origami structures with continuous constitutive curves, traditional parameter identification methods such as the least squares method[33] and subset selection[34] could effectively identify various physical parameters and reconstruct the dynamic model. However, static test results indicate that the constitutive curve of the KCO structure contains distinct discontinuities; therefore, we consider using a Gaussian-function-based piecewise system identification method[35] for model reconstruction. Based on the dynamic model, the bifurcation analysis with respect to the excitation amplitude is carried out. On the one hand, we will identify the different types of responses exhibited by the KCO structure; on the other hand, the transitions between these responses can also be clearly demonstrated. Additionally, considering the sensitivity of a nonlinear system to initial conditions, we

also performed a basin stability analysis. The result will reveal the evolution patterns of different types of responses and the probability of each response being triggered. **Overall, the innovations of this paper lie in taking the tri-stable KCO (with non-smooth piecewise characteristics) as the research object, establishing its simplified dynamic model, identifying its physical parameters using the Gaussian function-based method, and finally conducting a detailed study on its nonlinear dynamic characteristics.** This will provide a useful guideline for achieving dynamic control of the multi-stable origami structure.

The rest of this paper is organized as follows. Section II introduces the design and prototyping of the KCO structure, followed by the static test. In Section III, an equivalent dynamic model of the KCO structure is set up based on the piecewise nonlinear system identification method. Then, detailed dynamic analysis, including bifurcation analysis and basin stability analysis, is carried out in Section IV. Finally, the conclusion is given in Section V.

2 The Design of Kresling Conical Origami

2.1 The Geometry of the Kresling Conical Origami

Here, the Kresling conical origami (KCO) is chosen as the research object. In fact, Kresling conical origami is similar to the traditional Kresling origami. However, the difference is that Kresling origami is constructed from a series of repeated parallelogram units [29], [36], while the Kresling conical origami structure is composed of a series of quadrilaterals with a pair of opposite sides of equal length [37]. **Figure 1(a)** shows the crease pattern of a single-layer Kresling conical origami. It is composed of six quadrilateral units, each quadrilateral unit contains a valley folding crease line along one diagonal (blue dashed line) and two mountain folding crease lines (red solid lines). By overlapping points D and C with E and F , a single-layer triangulated conical origami could be obtained.

The Kresling conical origami has five design geometric parameters, which are the number of the quadrilateral units num , two angles α and β , the length l of the bottom edge $\overline{AD}$, the ratio ra of the length of the top edge $\overline{CB}$ and bottom edge $\overline{AD}$. Specifically, the value of ra needs to be less than 1. Based on this, the lengths of the folds in the crease pattern of **Fig.1(a)** are expressed as follows:

$$L_{DA} = l, L_{CB} = ra \cdot l, L_{CA} = \frac{l \sin(\alpha + \beta)}{\sin(\alpha)}, L_{CD} = L_{BA} = \frac{l \sin(\beta)}{\sin(\alpha)}. \quad (1)$$

In this research, the values of the five parameters are listed in **Table 1**. Many materials could be used to make the origami structure, considering both ease of fabrication and structural stiffness, 200 g art paper is selected as the raw material. Using the laser-based machining technology, the 2D crease pattern could be fabricated quickly. To reduce the bending stiffness of both mountain and valley folds, creases are perforated, and the whole structure becomes more flexible for folding. Small holes are also needed to cut at the vertices where multiple crease lines intersect. Furthermore, additional parts are also designed for connecting with the upper and bottom acrylic plates (as shown in **Fig.1(b)** and **(c)**).

Table 1. The design parameters of the Kresling conical origami.

Parameters	Value
l	35 mm
ra	2/3
num	6
α	38°
β	45°

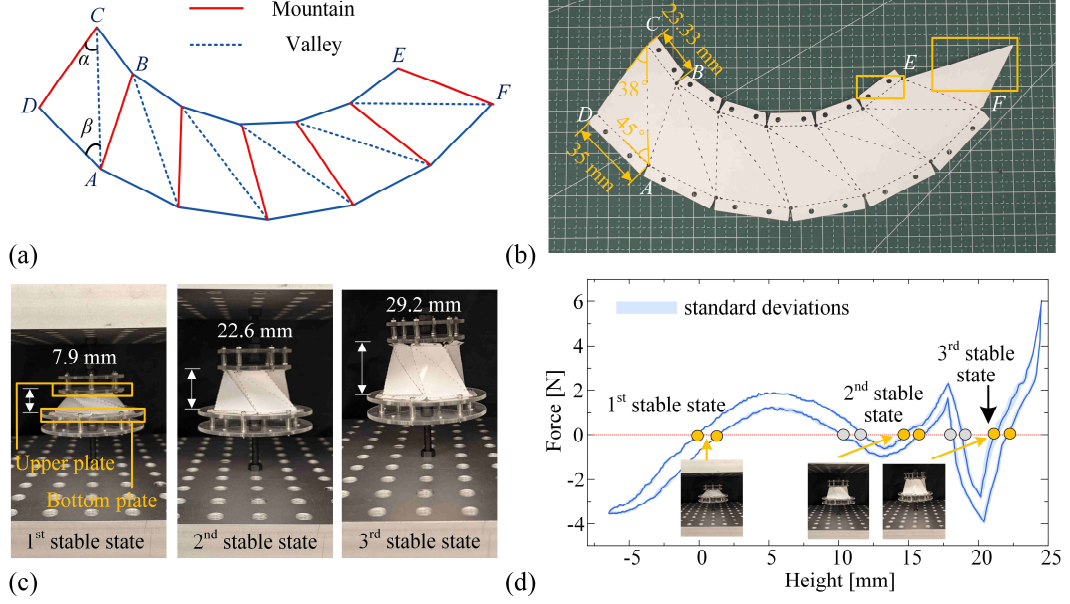


Fig. 1 Design, fabrication and testing of the Kresling conical origami. (a) The crease pattern of the Kresling conical origami. (b) The laser-machined 2D sheet of the Kresling conical origami is made of 200 g art paper. (c) The photo of the origami structure at three different stable states. (d) The experiment on the mechanical properties of the Kresling conical origami along the axial direction.

2.2 The static experiment

According to the research proposed by Wang et al [37], the Kresling conical origami structure based on the above design parameters (as listed in **Table 1**) could possess three stable states. As shown in **Fig.1(c)**, the heights of the 1st stable state, 2nd stable state, and 3rd stable state are 7.9 mm, 23.3 mm, and 29.2 mm, respectively. To obtain the constitutive relationship of the **triangulated** conical origami structure, the quasi-static tension and compression tests are performed. Using the screw, the origami structure could be fixed on the machine (**the tensile testing machine used in this study is INSTRON 5965, and the force sensor employed is INSTRON 2580-2KN**). The initial length of the Kresling conical origami is 7.9 mm (i.e., 1st stable state). We first compress it to 1.5 mm, and then stretch it to 32 mm, and finally compress it to its initial length. The loading speed is 0.25 mm/s, so the process can be ensured to be quasi-static. **Here, the deformation represents the change in height of the KCO structure. It could be calculated as the difference between the structural height and the height corresponding to the first stable state (i.e. 7.9 mm). Hence, the deformation of 0 mm corresponds to the structure being in the first stable state. The original data is shown in the Fig.1(d), We averaged the data from the three tests and marked the standard deviations.** It could be found that the force-displacement relationship curve has five intersections (three yellow circles and two grey circles in **Fig.1(d)**) with the x-axis, each corresponds to a stable configuration. **Then, the upper and lower branches of the hysteresis curve are averaged. The position of the first stable state on the averaged curve is translated to the origin, yielding the final result shown in Fig.2.** Furthermore, the curve also has two distinct breaks, i.e., two red circles shown in **Fig.2**. The positions of these two breaks are and (19.7, 0), respectively. To precisely tackle the nonlinear force-displacement relation and simplify the numerical simulation in the following section, we use the segmented polynomials to fit the experimental data.

$$f(x) = \begin{cases} p_{1,5}x^5 + p_{1,4}x^4 + p_{1,3}x^3 + p_{1,2}x^2 + p_{1,1}x + p_{1,0}, & x \leq 17.2 \text{ mm} \\ p_{2,1}x + p_{2,0}, & 17.2 \text{ mm} < x \leq 19.7 \text{ mm} \\ p_{3,3}x^3 + p_{3,2}x^2 + p_{3,1}x + p_{3,0}, & 19.7 \text{ mm} < x \end{cases} \quad (2)$$

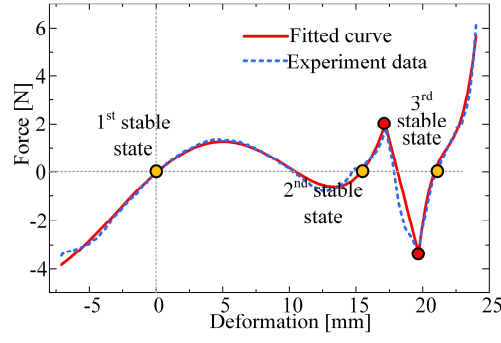


Fig. 2 The fitted curve and experimental data of the force-displacement relationship of the Kresling conical origami.

Table 2. The corresponding polynomial coefficients.

Coefficients	Value	Coefficients	Value
$p_{1,5}$	6.6353×10^{-6}	$p_{2,1}$	-2.1246
$p_{1,4}$	1.0209×10^{-4}	$p_{2,0}$	38.5073
$p_{1,3}$	-3.0906×10^{-3}	$p_{3,3}$	0.2731
$p_{1,2}$	-3.2016×10^{-2}	$p_{3,2}$	-17.8717
$p_{1,1}$	0.4817	$p_{3,1}$	390.6512
$p_{1,0}$	-4.5436×10^{-2}	$p_{3,0}$	-2851.4119

The corresponding polynomial coefficients are listed in **Table 2**. It can be seen that the correlation coefficient R^2 between the fitted curve and the experimental data is 0.993, indicating that the fitted curve captures the characteristics of the experimental data well. As a result, the numerical simulations in the following sections will be carried out based on this fitting model.

3 Methodology

3.1 Construction of the system

Similar to the Kresling structure, the KCO structure also undergoes torsional deformation when folded along the height direction, and the folded surfaces also exhibit in-plane deformation (i.e., non-rigid folding)[38]. However, in the specific dynamic analysis and experiments, we will only focus on its constitutive characteristic curve in the height direction and constrain any possible bending deformation. Meanwhile, an additional mass block will be added. Considering the lightweight nature of the origami structure itself, it can be equivalent to a nonlinear spring, and the entire KCO structure will be equivalent to a spring-mass oscillator system. The equivalent model is shown in **Fig. 3(a)**, and the external excitation $f(t)$ is applied at one end.

Furthermore, it should be noted that the units of mass, length, and time in this paper are kg, mm, and s, respectively. Therefore, the unit of constitutive force, which is N ($\text{kg} \cdot \text{m/s}^2$), will be converted to ($1000 \cdot \text{kg} \cdot \text{mm/s}^2$). In summary, the corresponding differential equation (i.e., equation of motion) fields

$$m\ddot{x} + c\dot{x} + 1000f_k(x) = f(t), \quad (3)$$

where c represents the equivalent viscous damping coefficient, $x(t)$ represents the global displacement of the deformation of the KCO structure. In the real experiment, the $x(t)$ and $f(t)$ actually are measured discretely at the time instants: $t_1, t_2, t_3, \dots, t_m$. The velocity $\dot{x}$ and the acceleration $\ddot{x}$ could be calculated by numerical methods, such as forward difference or backward difference. The $f_k(x)$ is the force-displacement curve of the origami structure. Considering its obviously piecewise behavior, we first give the universal expression as follows

$$f_k(x) = \begin{cases} g_1(x), d_0 < x < d_1, \\ \vdots \\ g_i(x), d_{i-1} < x < d_i, \\ \vdots \\ g_{r+1}(x), d_r < x < d_{r+1}, \end{cases} \quad (4)$$

where $r+1$ is the number of the segments, $g_i(x)$ ($i=1,2,3\dots r+1$) denotes an unknown nonlinear force-displacement in the i -th segment, which is composed of multiple basis functions. Hence, the Eq (1) could describe a piecewise nonlinear dynamic system. The system identification aims to determine the following key parameters: the positions of the non-smooth points (i.e., the value of d_i ($i=1,2,3\dots r$)) and the explicit expressions of $g_i(x)$ ($i=1,2,3\dots r+1$). Specifically, the value of d_0 and d_{r+1} need to be smaller or larger than the minimum and maximum of the output displacement, which could be determined by the experiment result. Additionally, physical parameters of the origami structure including mass m and damping coefficient c . For dimensional standardization, the units for mass, damping, stiffness, and displacement transition points are defined as kg, kg/s, N/mm, and mm, respectively. These unit specifications will not be reiterated in subsequent discussions to maintain conciseness.

3.2 Reconstruction of the EOM

To characterize the features of piecewise nonlinear systems while ensuring the smoothness and continuity of functions, we adopt the method proposed by Liu et al [35]. Firstly, the weighting functions $\varphi_i(x)$ are determined, they should form a partition of unity and be normalized as

$$\sum_{i=1}^{r+1} \varphi_i(x) = 1, \quad (5)$$

where x is the independent variable of the piecewise function. It should be noted that each $\varphi_i(x)$ has a value of 1 only in the corresponding region (i.e., $[d_{i-1}, d_i]$) and a value of 0 in all other regions. On the other hand, in order to ensure the continuity of the function, the specific form of $\varphi_i(x)$ is as follows.

$$\varphi_i(x) = \frac{G_i(x)}{\sum_{j=1}^{r+1} G_j(x)}, \quad (6)$$

where $G_i(x)$ is given as follows

$$G_i(x) = e^{-\frac{(x-\mu_i)^2}{2\sigma_i^2}}. \quad (7)$$

In this equation, μ_i represents the expected value, σ_i represents the standard deviation quantifying the degree of variation. To ensure that only a single weighting function $\varphi_i(x)$ in each interval $[d_{i-1}, d_i]$ is activated, the parameters are designed as follows: the μ_i is located at the center of the interval $[d_{i-1}, d_i]$, while the value of the σ_i needs to be close to zero

$$\mu_i = \frac{d_{i-1} + d_i}{2}, \sigma_i = \gamma(d_{i-1} - d_i). \quad (8)$$

γ is a proportionality factor used to relate the standard deviation of the validity function to the extension range of the region. The value of γ should be kept as small as possible to ensure that σ_i approaches zero.

Based on the weighting function mentioned above, the equation of motion **Eq.(3)** could be rewritten as

$$m\ddot{x} + c\dot{x} + 1000 \sum_{i=1}^{r+1} g_i(x) \varphi_i(x) = f(t). \quad (9)$$

In fact, many experiments could only obtain the discrete time history data of $x(t)$. To further calculate the corresponding velocity $\dot{x}(t)$ and acceleration $\ddot{x}(t)$, the B-spline Galerkin integration method is employed [39]. This method could not only avoid amplifying noise but also ensure the continuity of data. The cubic B-splines are used as the basis functions $B_j(t)$ ($j=1,2,\dots,p$), and the excitation $f(t)$ and displacement response $x(t)$ could be rewritten as

$$f(t) = \sum_{j=1}^p a_j B_j(t), x(t) = \sum_{j=1}^p b_j B_j(t), \quad (10)$$

where the number p of the basis functions equals the number of sampling points M , so that any degree of change in the data can be captured by the expressions. Although excessive basis functions may introduce noise interference, the proposed integration algorithm could effectively suppress such errors [35]. The specific values of coefficients a_j and b_j can be calculated by substituting the experimental data $x(t)$ and $f(t)$ into **Eq. (10)**.

Substituting **Eq. (10)** into the equation of motion **Eq. (9)**, and simultaneously integrating both ends from t_1 to t_m . Then, using the integration by parts and the boundary condition that $B_j(t)$, $\dot{B}_j(t)$ are equal to 0 at both ends t_1 and t_m , the above-mentioned equation could be transformed into

$$\int_{t_1}^{t_m} B_j(t) \left(m\ddot{x} + c\dot{x} + 1000 \sum_{i=1}^{r+1} g_i(x) \varphi_i(x) \right) dt = \int_{t_1}^{t_m} B_j(t) f(t) dt. \quad (11)$$

To simplify the expression, three integral operators could be defined, and Eq. (11) could be rewritten as

$$\begin{aligned} \alpha_j(mx) + \beta_j(cx) + 1000 \sum_{i=1}^{r+1} \chi_j(g_i(x) \varphi_i(x)) &= \chi_j(f(t)), \\ \alpha_j(y(t)) &= \int_{t_1}^{t_m} \ddot{B}_j(t) y(t) dt, \quad \beta_j(y(t)) = \int_{t_1}^{t_m} \dot{B}_j(t) y(t) dt, \\ \chi_j(y(t)) &= \int_{t_1}^{t_m} B_j(t) y(t) dt. \end{aligned} \quad (12)$$

3.3 Construction of the identification method

Based on the above equation, the identification method could be constructed as shown in **Fig.3(c)**.

- I. In this method, only the number of discontinuity points is known, while their specific locations remain undetermined. Given the form of $g_i(x) = p_{i,1}h_1(x) + p_{i,2}h_2(x) + \dots + p_{i,n_i}h_{n_i}(x)$, where $h_k(x)$ is known and represents elementary functions (e.g. $x^2, x^3, \sin(x), etc$), p_{i,n_i} is unknown and represents the coefficient which is the object to be identified.
- II. In the actual experimental tests or simulation experiments, both the external excitation input $f(t)$ and the corresponding response $x(t)$ are known, and we possess their time-history data. Therefore, they will serve as the inputs to the identification algorithm.
- III. The initial value of the non-smooth positions (i.e., d_i^1 ($i=1,2,3\dots r$)) and the starting/ending displacement (i.e. d_0 and d_{r+1}) need to be determined. The value of d_0 needs to be smaller than the minimum of $x(t)$, the value of d_{r+1} needs to be larger than the maximum of $x(t)$.

The initial value of the d_i^l needs to follow a certain increasing sequence, i.e., $d_0 < d_1 < \dots < d_r < d_{r+1}$.

IV. By substituting the time-history data of $f(t)$ and $x(t)$ into **Eq. (12)**, it can be rewritten as:

$$\alpha_j(mx) + \beta_j(cx) + 1000 \sum_{i=1}^{r+1} \chi_j \left((p_{i,1}h_1(x) + p_{i,2}h_2(x) + \dots + p_{i,n_i}h_{n_i}(x)) \varphi_i(x) \right) = \chi_j(f(t)). \quad (13)$$

According to the **Eq (12)**, all coefficients can be factored out, and the equation can be further written as:

$$m\alpha_j(x) + c\beta_j(x) + 1000 \sum_{i=1}^{r+1} \left(p_{i,1}\chi_j(h_1(x)\varphi_i(x)) + p_{i,2}\chi_j(h_2(x)\varphi_i(x)) + \dots + p_{i,n_i}\chi_j(h_{n_i}(x)\varphi_i(x)) \right) = \chi_j(f(t)). \quad (14)$$

As shown in the **Eq (14)**, the parameters $m, c, p_{1,1}, p_{1,2}, \dots, p_{1,n_1}, p_{r+1,1}, p_{r+1,2}, \dots, p_{r+1,n_{r+1}}$ are linear, and the **EOM** could be further degenerated into

$$\mathbf{X}\boldsymbol{\theta} = \mathbf{F}, \quad (15)$$

where $\mathbf{X}$, $\boldsymbol{\theta}$, and $\mathbf{F}$ field

$$\mathbf{X} = \begin{bmatrix} \alpha_1(x) & \beta_1(x) & \chi_1(h_1(x)\varphi_1(x)) & \chi_1(h_2(x)\varphi_1(x)) & \cdots & \chi_1(h_{n_1}(x)\varphi_1(x)) \\ \alpha_2(x) & \beta_2(x) & \chi_2(h_1(x)\varphi_1(x)) & \chi_2(h_2(x)\varphi_1(x)) & \cdots & \chi_2(h_{n_1}(x)\varphi_1(x)) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \alpha_M(x) & \beta_M(x) & \chi_M(h_1(x)\varphi_1(x)) & \chi_M(h_2(x)\varphi_1(x)) & \cdots & \chi_M(h_{n_1}(x)\varphi_1(x)) \\ \chi_1(h_1(x)\varphi_2(x)) & \chi_1(h_2(x)\varphi_2(x)) & \cdots & \chi_1(h_{n_2}(x)\varphi_2(x)) \\ \chi_2(h_1(x)\varphi_2(x)) & \chi_2(h_2(x)\varphi_2(x)) & \cdots & \chi_2(h_{n_2}(x)\varphi_2(x)) \\ \vdots & \vdots & \vdots & \vdots \\ \chi_M(h_1(x)\varphi_2(x)) & \chi_M(h_2(x)\varphi_2(x)) & \cdots & \chi_M(h_{n_2}(x)\varphi_2(x)) \\ \chi_1(h_1(x)\varphi_{r+1}(x)) & \chi_1(h_2(x)\varphi_{r+1}(x)) & \cdots & \chi_1(h_{n_{r+1}}(x)\varphi_{r+1}(x)) \\ \chi_2(h_1(x)\varphi_{r+1}(x)) & \chi_2(h_2(x)\varphi_{r+1}(x)) & \cdots & \chi_2(h_{n_{r+1}}(x)\varphi_{r+1}(x)) \\ \vdots & \vdots & \vdots & \vdots \\ \chi_M(h_1(x)\varphi_{r+1}(x)) & \chi_M(h_2(x)\varphi_{r+1}(x)) & \cdots & \chi_M(h_{n_{r+1}}(x)\varphi_{r+1}(x)) \end{bmatrix}, \quad (16a)$$

$$\boldsymbol{\theta} = [m, c, p_{1,1}, \dots, p_{1,n}, p_{2,1}, \dots, p_{2,n}, p_{r+1,1}, \dots, p_{r+1,n}]^T, \quad (16b)$$

$$\mathbf{F} = [\chi_1(f(t)), \chi_2(f(t)), \chi_3(f(t)), \dots, \chi_M(f(t))]^T. \quad (16c)$$

The parameters vector $\boldsymbol{\theta}$ could be calculated by

$$\boldsymbol{\theta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{F}. \quad (17)$$

V. After calculating $\boldsymbol{\theta}$, we compute the gradient corresponding to the current discontinuity points and update these points. Specifically, the gradient of each non-smooth point d_i^k (k denotes the iteration count during the update process) is given by

$$\nabla J_i = \sum_{l=1}^{r+1} C_j [g_l(x) \frac{\partial \varphi_l(x)}{\partial d_i^k}]. \quad (18)$$

Then, the value of d_i^k could be updated to d_i^{k+1} based on the gradient descent method

$$d_i^{k+1} = 0.5d_i^k + \alpha \cdot \nabla J_i, \quad (19)$$

where α is the learning rate. The cut-off condition for the identification algorithm iteration is

$$\sum_{i=1}^r |d_i^{k+1} - d_i^k| < error_1 \quad (20)$$

where $error_1$ is the cut-off error for the whole identification algorithm.

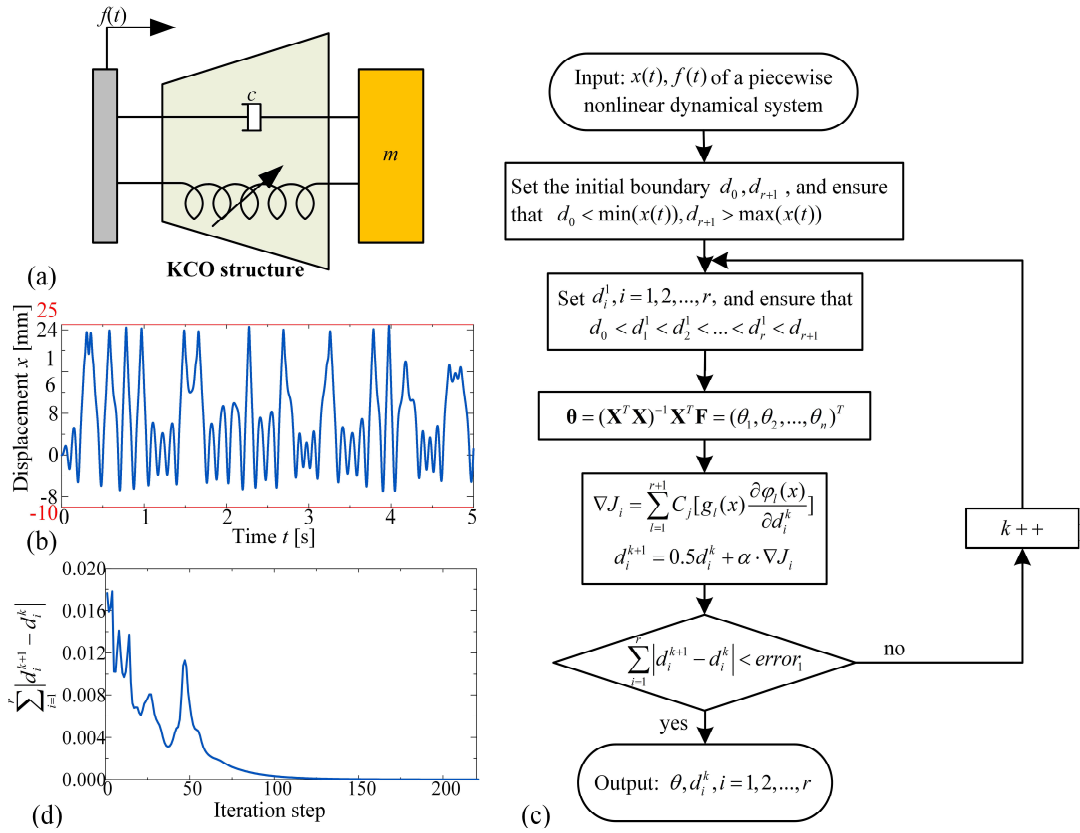


Fig. 3 The parameter identification algorithm and dynamic system of the KCO. (a) The equivalent model of the KCO structure. (b) The noisy displacement response signal x with S/N = 60 dB. (c) The flow diagram of the parameter identification algorithm. (d) The variation of $\sum_{i=1}^r |d_i^{k+1} - d_i^k| < error_1$ with the iteration process.

Based on the above identification method, the numerical simulation experiment could be carried out. Since a displacement excitation is applied, the EOM in Equation (3) further evolves into

$$\begin{aligned} m\ddot{x} + c\dot{x} + 1000f_k(x) &= f(t), f(t) = -m\ddot{u}_{in}, \\ x &= u_{out} - u_{in}, u_{in} = u \sin(\omega t), \end{aligned} \quad (21)$$

where m and c denote the mass and viscous damping, with values of 0.1 kg and 0.05 kg/s, respectively. This set of parameters is selected to ensure that the subsequent KCO dynamic simulation exhibits rich dynamic responses. u_{in} is the displacement excitation applied on the left side of the KCO structure (as shown in Fig.3(a)), while u_{out} is the displacement of the mass. The form of u_{in} is $u \sin(\omega t)$, with amplitude u (1.05 mm) and frequency ω (20π rad/s). x is the axial deformation of the KCO structure. This external excitation parameter ensures that the KCO undergoes

Table 3. The corresponding polynomial coefficients.

Parameters	True Value	Identified Value	Relative Error
d_1	17.2 mm	17.19 mm	0.06 %
d_2	19.7 mm	19.71 mm	0.05 %
m	0.1 kg	0.097 kg	3.00 %
c	0.05 kg/s	0.051 kg/s	2.00 %
$p_{1,5}$	6.6353×10^{-6}	6.5454×10^{-6}	1.35 %
$p_{1,4}$	1.0209×10^{-4}	0.9912×10^{-4}	2.99 %
$p_{1,3}$	-3.0906×10^{-3}	-3.0008×10^{-3}	2.90 %
$p_{1,2}$	-3.2016×10^{-2}	-3.1047×10^{-2}	3.03 %
$p_{1,1}$	0.4817	0.4681	2.82 %
$p_{1,0}$	-4.5436×10^{-2}	-4.4035×10^{-2}	3.08 %
$p_{2,1}$	-2.1246	-2.0914	1.56 %
$p_{2,0}$	38.5073	37.3659	2.97 %
$p_{3,3}$	0.2731	0.2655	2.79 %
$p_{3,2}$	-17.8717	-17.3735	2.79 %
$p_{3,1}$	390.6512	379.7605	2.79 %
$p_{3,0}$	-2851.4119	-2772.0503	2.79 %

large-amplitude inter-well vibration. For this parameter identification method, the input time-history data must contain the global nonlinear information of the KCO; therefore, small-amplitude intra-well vibration will affect the identification accuracy. The 4th order Runge-Kutta method is applied to solve the Eq. (14) numerically. The simulation duration is 5 seconds, with a sampling interval of 0.005 seconds. We also add Gaussian white noise with S/N=60 dB to the displacement response signal x (the effect of the S/N would be discussed in Appendix A).

During the identification process, we only know that the function $f_k(x)$ has two non-smooth points, but their locations (i.e. d_1 and d_2) need to be determined; in addition, the polynomial forms on each segment of $f_k(x)$ are determined, but the value of the corresponding polynomial coefficients need to be determined. Firstly, we determine the values of d_0 and d_3 based on the time-history data of x obtained from the numerical simulation (as shown in Fig.3(b)). d_0 should be less than the minimum value of x and is set to -10 here; d_3 should be larger than the maximum value of x and is set to 25 here. The proportionality factor γ is set to be 1/30, the cut-off error $error_1$, respectively. The initial guessed value of the d_1 and d_2 are set to 18 and 20, respectively. Then, based on the algorithm shown in Fig.3(c), the iterative parameter identification process could begin. Fig.4(c) shows the value of $\sum_2 |d_i^{k+1} - d_i^k|$ with respect to the iteration steps, it could observed that It can be observed that after 47 iterations, this value begins to converge rapidly. After 220 iterations, the algorithm meets the cut-off condition, and the identified parameters are presented in Table 3. The obtained identified parameters are very close to the true value, the relative errors are all smaller than 3.08%. It can be concluded that this method could accurately reconstruct the dynamic model of the KCO structure, and following studies will be carried out based on the identified parameters.

To further verify the accuracy of the aforementioned identification method, we will analyze it from both simulation and experimental perspectives herein. Firstly, a numerical simulation is carried out for both the initial parameters and the identified parameters. The input amplitude is 0.8 mm, while the input frequency is 9 Hz. The results are shown in Fig.4(a) and (b), including the displacement-time graph and velocity-time graph from 45 to 50 seconds. In the figure, the blue solid line represents the simulation results of the initial parameters, and the red dashed line represents the simulation results of the identified data. By analyzing the coefficient of determination R2 of the two, we can see that the values are 0.988 and 0.985, respectively. This indicates that the two sets of simulation results are very close, which also demonstrates the reliability of the subsequent dynamic simulation results.

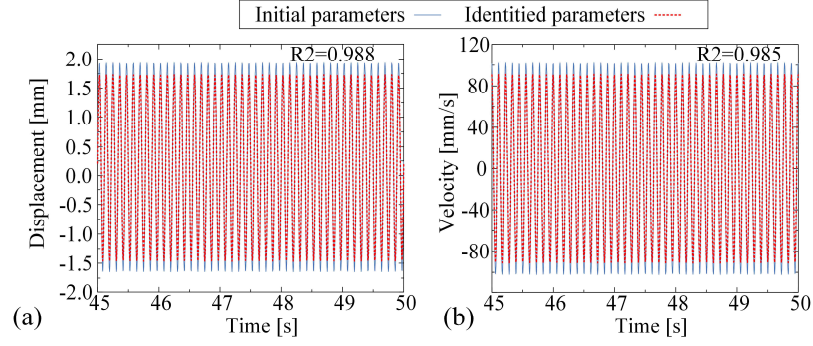


Fig. 4 Analysis of the accuracy of dynamic simulations based on identified parameters: (a) displacement, (b) velocity.

4 Dynamic Analysis

4.1 Bifurcation Analysis

Based on the dynamic model established in the above section, a comprehensive dynamic analysis is carried out. For multi-stable origami structures (e.g., Stacked Miura-ori cell), their rich dynamic responses can be divided into two categories, intra-well and inter-well. For the former, the vibration only occurs near one of the stable configurations, while for the latter, the origami structure would snap between multiple stable configurations. For our KCO structure, there are three stable configurations in total, so it has richer dynamic responses.

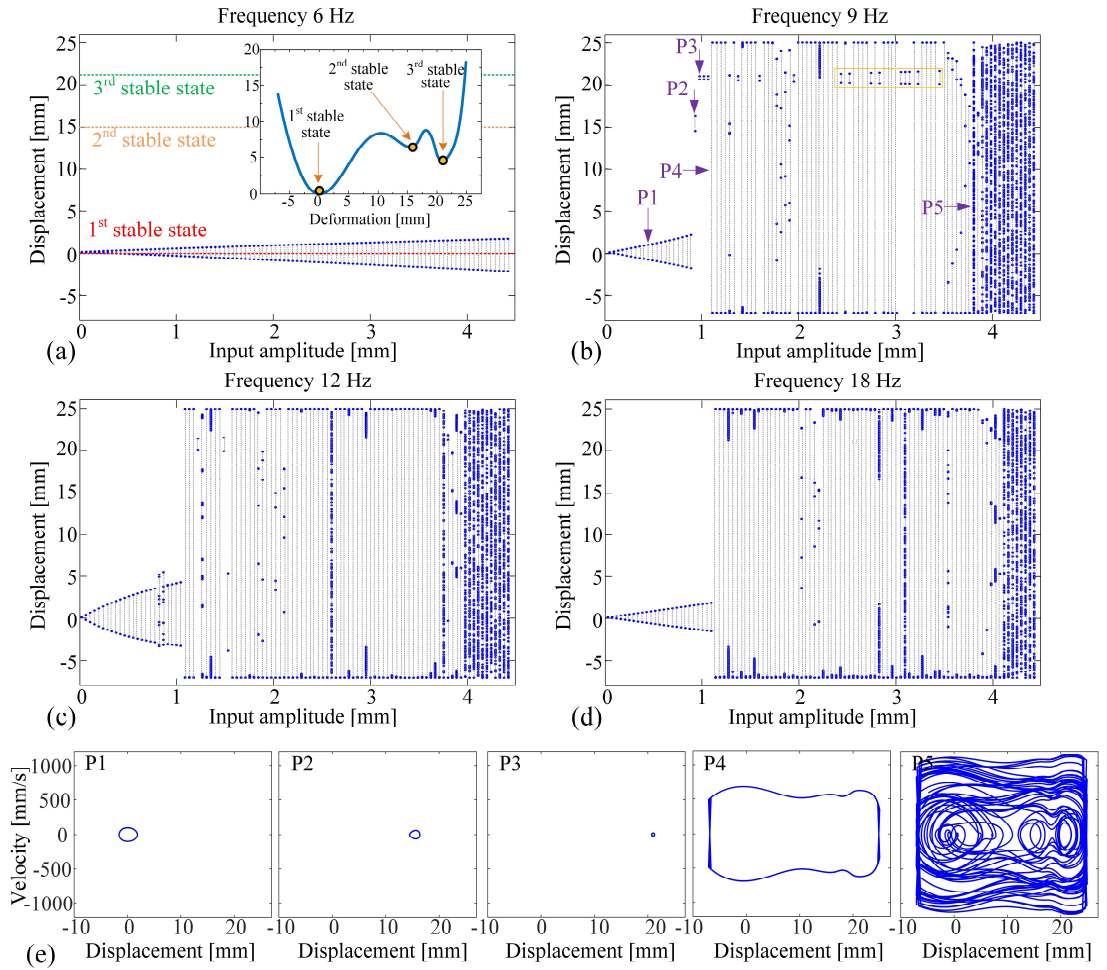


Fig. 5 Bifurcation diagrams of the KCO structure. (a) 6 Hz, (b) 9 Hz, (c) 12 Hz, and (d) 18 Hz. In each case, the amplitude of the input ranges from 0.005 mm to 4.5 mm. The maximum and minimum displacements of the KCO structure are plotted for each of the 25 cycles after the system reaches the steady state. (e) Phase diagrams of the KCO structure at five input amplitudes.

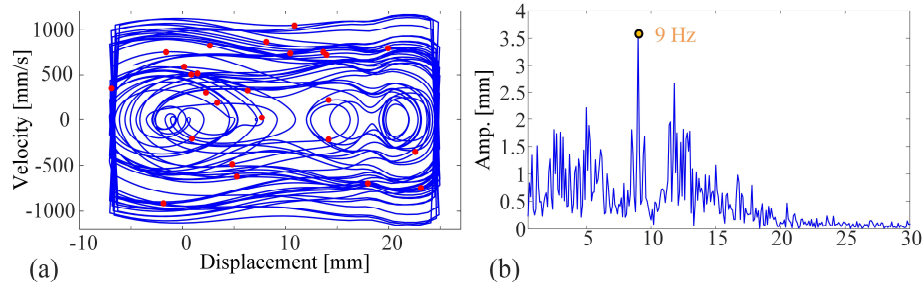


Fig. 6 The chaos analysis of the dynamic response corresponding to point P5. (a) Poincaré section. (b) FFT spectra.

Here, we first analyze how the steady state response of the KCO structure changes as the amplitude of the input changes. For the actual KCO structure, there are displacement constraints under both fully extended and compressed states. To prevent structural damage to the KCO structure during simulation, we drew inspiration from the work of Han et al[40] and added additional restoring forces at both ends of the constitutive curve (the effect of the constraint would be analyzed in Appendix B). Then, the Eq (2) yields

$$f_k(x) = \begin{cases} 100000(x+7.1) - 3.8333, & x \leq -7.1 \text{ mm} \\ \sum_{i=1}^3 g_i(x) \phi_i(x), & -7.1 \text{ mm} < x \leq 25 \text{ mm} \\ 100000(x-25) + 12.477, & 25 \text{ mm} < x \end{cases} \quad (22)$$

where the corresponding parameters are listed in Table 3. The external excitation frequency is selected as 6 Hz, 9 Hz, 12 Hz, and 18 Hz, respectively. The displacement of the excitation is swept from 0.005 mm to 4.5 mm. For each amplitude, 1000 times of the excitation period is set for simulation to reach the state. It should be noted here that in this bifurcation analysis, the initial condition for all simulations is set as the KCO system being at the first equilibrium state with an initial velocity of 0. The influence of initial conditions on the dynamic response of the KCO will be analyzed in Section 4.2.

The simulation result is shown in Fig.5. At each input amplitude, the maximum and minimum displacements are plotted for the last 25 periods after the system reaches the steady state. In Fig.5(a), the location of three stable states is indicated by red, green, and yellow dashed lines, respectively. When the input frequency is 6 Hz, the KCO structure undergoes intra-well oscillation near the first stable state as the input amplitude increases from 0.005 mm to 4.5 mm, and the response amplitude gradually increases from 0 mm to 3.9 mm. This is because the energy of the external excitation is insufficient for the KCO structure to overcome its potential energy barrier. In the case of a 9 Hz excitation frequency, the KCO structure exhibits more diverse dynamic responses. We selected five amplitude values and plotted the corresponding dynamic response phase diagrams, as shown in Fig.5(e). When the input amplitude is small, such as 0.414 mm (P1), the KCO undergoes period-1 intra-well oscillation around the first stable state. As the input amplitude increases to 0.913 mm (P2), the KCO structure switches from the first stable state to the second stable state and performs intra-well oscillations around the second stable state. As the excitation amplitude further increases to 0.958 mm (P3), the KCO switches to the third stable point and performs periodic intra-well oscillation. Keep increasing the amplitude, and the period-1 inter-well oscillation among three stable states would be triggered at 1.095 mm (P4). Due to the strong nonlinearity of the KCO structure, it is highly sensitive to initial conditions. Therefore, even when the excitation amplitude exceeds the P4 point, it is still possible for the system to exhibit intra-well oscillations near a certain stable state (e.g., as

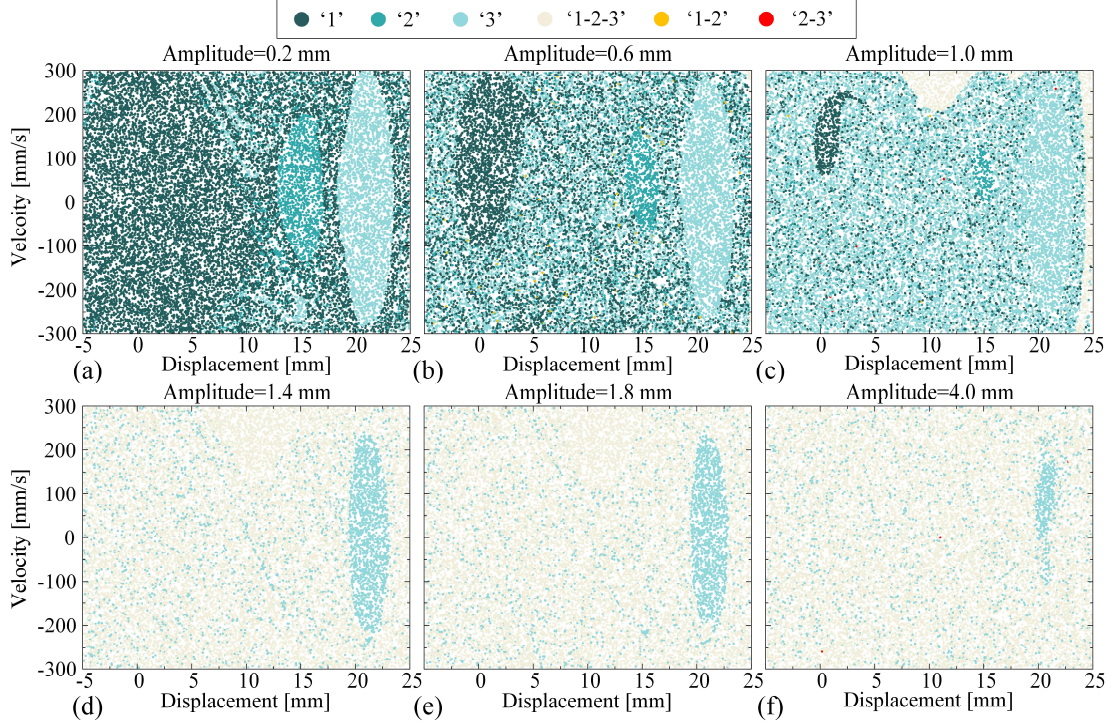


Fig. 7 Basins of attraction on the displacement-velocity plane corresponding to 6 excitation amplitudes when the frequency is 9 Hz. (a) Amplitude is 0.2 mm, (b) amplitude is 0.6 mm, (c) amplitude is 1.0 mm, (d) amplitude is 1.4 mm, (e) amplitude is 1.8 mm, (f) amplitude is 4.0 mm. The 6 different types of dynamic responses are shown by different colors.

shown in the yellow box in **Fig.5(b)**, intra-well oscillation near the third stable state). Finally, when the input amplitude reaches 3.864 mm (i.e., P5 point), a chaotic response is triggered. **To confirm that the response is chaotic, we plotted the Poincaré section and the frequency response curve as shown in the Fig.6. The Poincaré section shows dense points (red points), and a continuous frequency band is observed in the FFT spectra. These two pieces of evidence confirm that the dynamic response at point P5 is chaotic.** At excitation frequencies of 12 Hz and 18 Hz, the KCO structure also exhibits various dynamic responses as the excitation amplitude increases, including period-doubling responses.

It should be noted here that the evolution process shown in **Fig.7(b)** can actually be further explained based on the potential energy curve of the KCO structure. The potential energy landscape is shown in the **Fig.5(a)**. It can be observed that the depths of the three potential energy wells, from deepest to shallowest, are in the order of the first stable state ($x_{\text{stable-1}} = 0$ mm), the third stable state ($x_{\text{stable-3}} = 20.89$ mm), and the second stable state ($x_{\text{stable-2}} = 15.43$ mm). In fact, this phenomenon can also explain, to a certain extent, the evolution process shown in Fig.5(b). Under small-amplitude conditions, since the initial condition of the KCO structure is located at the first stable state, it undergoes intra-well vibrations near the first stable state over a large range of external excitation amplitude variations. As the external excitation amplitude increases to P2, the second steady state is close to the first steady state, leading to transient intra-well oscillations near the second stable state. However, as the external excitation amplitude continues to increase, the system immediately switches to intra-well vibrations near the third stable state.

4.2 Basin Stability

In the bifurcation analysis above, we observed that the KCO structure exhibits a variety of dynamic responses even as the excitation amplitude continuously increases. Considering the extreme sensitivity of nonlinear systems to initial conditions, we introduce the concept of basin stability to quantify the probability of different dynamic behaviors. Menck et al.[41] used the concept of basin

stability to characterize the ‘stability’ of a stable attractor, which is quantified by the basin’s volume. Based on the characteristics of the displacement response amplitude, there are six different dynamic response types: intra-well (‘1’) responses (i.e., around the first stable state), intra-well (‘2’) responses (i.e., around the second stable state), intra-well (‘3’) responses (i.e., around the third stable state), inter-well (‘1-2-3’) responses (i.e., around all three stable states), inter-well (‘1-2’) responses (i.e., around first and second stable states), and inter-well (‘2-3’) responses (i.e., around second and third stable states).

In this case, the excitation frequency is set to 9 Hz, and the amplitudes are chosen as 0.2 mm, 0.6 mm, 1 mm, 1.4 mm, 1.8 mm, and 4.0 mm. For each excitation amplitude, 20000 initial conditions are selected randomly from the displacement-velocity plane $\text{displacement} \in [-5, 25] \text{ mm} \times \text{velocity} \in [-300, 300] \text{ mm/s}$. Similar to the previous setup, for each initial condition, the simulation time is set to 1000 times the excitation period to reach a steady state. **The classification criteria for dynamic response types are defined based on their inherent characteristics. For example, regarding the dynamic response ‘1’—which refers to inter-well vibrations around the first stable state—the maximum value of its displacement time history must be less than the position corresponding to the second equilibrium point ($x_{\text{stable-2}}$) and the minimum value larger than the position corresponding to the first equilibrium point ($x_{\text{stable-1}}$) i.e., $\max(\mathbf{x}_{\text{final}}) < x_{\text{stable-2}}, \min(\mathbf{x}_{\text{final}}) < x_{\text{stable-1}}$. This ensures that the displacement time history only includes the first stable state. The judgment criteria for the remaining responses are presented in the Table 4.**

Figure 7 shows the evolution of six dynamic response types, and their respective proportions (i.e., basin stability) are listed in **Table 5**. When the input amplitude is only 0.6 mm, the injected energy is small, and only intra-well responses could be observed. Due to the differences in the width and depth of the potential wells associated with the three stable states of the KCO structure, the probabilities of the three types of intra-well responses also vary. Among them, ‘1’ intra-well response has the highest probability at 70.98%, while ‘2’ has the lowest at 11.73%. As the amplitude increases to 1.0 mm, inter-well responses of types ‘1-2-3’, ‘1-2’, and ‘2-3’ all emerge. Moreover, the occurrence probability of ‘1’ intra-well response decreases to 13.86%, while that of ‘3’ inter-well response increases to 70.13%. After the input amplitude reaches 1.4 mm, ‘3’ intra-well response and ‘1-2-3’ inter-well response become dominant. Furthermore, as the amplitude continues to increase, the probability of ‘1-2-3’ inter-well response continues to rise. When the amplitude reaches 4 mm, this

Table 4 The design parameters of the Kresling conical origami.

Dynamic Response	Criteria for Classification
“1”	$\max(\mathbf{x}_{\text{final}}) < x_{\text{stable-2}}, \min(\mathbf{x}_{\text{final}}) < x_{\text{stable-1}}$
“2”	$\max(\mathbf{x}_{\text{final}}) < x_{\text{stable-3}}, \min(\mathbf{x}_{\text{final}}) > x_{\text{stable-1}}$
“3”	$\max(\mathbf{x}_{\text{final}}) > x_{\text{stable-3}}, \min(\mathbf{x}_{\text{final}}) > x_{\text{stable-2}}$
“1-2-3”	$\max(\mathbf{x}_{\text{final}}) > x_{\text{stable-3}}, \min(\mathbf{x}_{\text{final}}) < x_{\text{stable-1}}$
“1-2”	$\max(\mathbf{x}_{\text{final}}) < x_{\text{stable-3}}, \min(\mathbf{x}_{\text{final}}) < x_{\text{stable-1}}$
“2-3”	$\max(\mathbf{x}_{\text{final}}) > x_{\text{stable-3}}, \min(\mathbf{x}_{\text{final}}) > x_{\text{stable-1}}$

Table 5. The basin stability of different input amplitudes.

Input Amplitude	‘1’	‘2’	‘3’	‘1-2-3’	‘1-2’	‘2-3’
0.2 mm	70.98%	11.73%	17.29%	0%	0%	0%
0.6 mm	41.90%	11.91%	45.57%	0.18%	0.43%	0%
1.0 mm	13.86%	6.62%	70.13%	9.33%	0.02%	0.04%
1.4 mm	0%	0%	19.93%	80.07%	0%	0%
1.8 mm	0%	0%	16.86%	83.14%	0%	0%
4.0 mm	0%	0%	12.92%	87.07%	0%	0.01%

probability reaches 80%, gradually spreading to dominate globally. The above results of the basins of attraction and their stability clearly demonstrate that the probabilities of different types of dynamic responses are highly related to the input amplitude. This analysis also provides a basis for controlling the dynamic responses of the KCO structure, thereby enhancing its dynamic performance.

5 Conclusion

To further advance the study of multi-stable origami structures from the field of statics to dynamics, this paper selects the KCO as the research platform and systematically performs analyses covering fabrication, static testing, parameter identification modeling, and dynamic analysis.

Different from the Kresling origami structure, the KCO (Kresling conical origami) structure can achieve tri-stability through the precise design of geometric parameters. Specifically, the KCO structure in our research is made of 200 g of art paper. Axial compression test indicates that the KCO structure possesses three stable points and two unstable points. Noting that its force-displacement curve exhibits a distinct piecewise characteristic, we use piecewise polynomials for fitting.

To deepen the understanding of the dynamics of the KCO structure, a dynamic model of the KCO structure is developed by employing a nonlinear system identification method. The whole KCO structure could be equivalently modeled as a nonlinear spring-damping system. Then, a Gaussian function-based piecewise nonlinear system identification method is adopted. Using noise-added simulation data as input, the method can accurately identify parameters including mass, damping, segment points of the force-displacement curve, and the polynomial expressions for each segment.

Based on the identified model, a comprehensive dynamic analysis is carried out. The bifurcation analysis of external excitation amplitude and frequency reveals that the KCO structure exhibits rich dynamic response types. When the excitation frequency is relatively low (i.e., 6 Hz), the KCO structure maintains small-amplitude intra-well oscillations. As the excitation frequency increases to 9 Hz, the KCO structure sequentially undergoes intra-well oscillations around the three stable states with increasing amplitude, eventually triggering large-amplitude inter-well oscillations involving all three stable states.

We also study the effect of the initial condition in terms of the basins of attraction and the basin stability. The former enables the analysis of how different dynamic responses evolve as the input amplitude varies. For the latter, the probability of various dynamic responses can be quantitatively evaluated. It is observed that only intra-well responses are present in the KCO structure when the amplitude of the input is small; as the amplitude increases, inter-well responses around the three stable states become dominant.

Overall, the results of this study provide a complete framework—from fabrication and modeling to dynamic analysis—for investigating the dynamics of piecewise nonlinear multistable origami structures. It lays a solid foundation for future studies on dynamic regulation. **Future research can be carried out in several aspects: 1) The geometric parameters of the KCO affect its multi-stability characteristics. For example, under extreme conditions, it can degenerate into a bistable Kresling structure[37]. Therefore, further analysis and exploration of geometric parameters can be conducted in subsequent studies, including the effect of the environment. 2) The damping model in this paper needs further improvement. Under actual working conditions, the KCO may be affected by factors including viscous friction and Coulomb dry friction, requiring a more accurate and complex friction model. 3) Whether there are corresponding conditions for triggering the various dynamic responses discovered in this paper, and whether there are differences in the probability of occurrence of different dynamic responses. 4) Intelligent materials can be considered for introduction to realize active control of the dynamic response of the KCO structure.**

Appendix A

The setting of the stopping tolerance error_1 needs to be balanced between calculation accuracy and efficiency. Theoretically, this parameter can take any positive value, and the smaller the value, the stricter the convergence criterion, and the higher the accuracy of the obtained solution. In this study, it is set to 1×10^{-6} , aiming to ensure that the parameter identification results have a high numerical accuracy. The optimization settings of the learning rate is shown in the following table. The first two rows correspond to the learning rates for two discontinuity points, followed by the number of iterations and the coefficients to be identified, and the meaning of the last row error_1 is "whether the iteration stop condition is met". It can be observed that setting the learning rate too large is likely to cause the algorithm to oscillate and fail to converge, while setting it too small will significantly increase the number of iterations and reduce the calculation efficiency.

Table A1 The effect of the learning rate

α_1	0.00003	0.00003	0.00001	0.00001	0.000001	0.000001
α_2	0.00003	-0.00003	-0.00001	0.00001	-0.000001	0.000001
<i>iterations</i>	10	600	600	600	21	600
m	NaN	38.3603237	62.7012962	61.9656396	97.2738652	12.2038693
c	NaN	63.1823701	57.9221212	58.0833576	50.5364116	68.8934003
$p_{1,5}$	NaN	-15.404664	4.43236696	5.5646131	6.46270613	2.96567244
$p_{1,4}$	NaN	126.569486	64.2188206	45.987303	99.5529848	-11.962334
$p_{1,3}$	NaN	-230.88195	-2058.7046	-2063.2367	-3015.5257	-677.09557
$p_{1,2}$	NaN	-14760.474	-19816.968	-18785.112	-31135.091	-2159.5113
$p_{1,1}$	NaN	184926.155	310853.627	307462.02	469207.015	79155.9839
$p_{1,0}$	NaN	-7899.1177	-28531.36	-32665.521	-44152.652	-12276.23
$p_{2,1}$	NaN	-117495.95	763977.945	488472.464	-2068410.4	11041.454
$p_{2,0}$	NaN	1288248.25	-11961247	-7430369.8	37488220.8	-199887.28
$p_{3,3}$	NaN	20111.4968	-23669.201	-24091.759	266102.066	2137741.73
$p_{3,2}$	NaN	-1103237.5	1752188.47	1776476.24	-17413866	165178150
$p_{3,1}$	NaN	19908543.2	-41485625	-41952018	380664065	-1.211E+10
$p_{3,0}$	NaN	- 118220202	317043302	320043011	- 2.779E+09	1.6601E+11
d_1	NaN	6.1105	17.3	21.0067	17.2006	14.0915
d_2	NaN	14.2436	16.3732	11.7727	19.7011	24.6411
" error_1 "	no	no	no	no	yes	no

α_1	0.0000001	0.0000001	0.00000015	0.0000002	0.0000001	0.00000001
α_2	-0.0000001	-0.0000001	-1.5E-07	-0.0000002	-0.0000002	-1E-08
<i>iterations</i>	231	600	158	119	218	1000
m	97.2738656	13.7223551	97.2738692	97.2738662	97.2738609	97.2731412
c	50.5364096	68.8657196	50.5364094	50.5364105	50.536417	50.533845
$p_{1,5}$	6.46270671	1.56825192	6.46270671	6.46270645	6.4627071	6.4617892
$p_{1,4}$	99.552974	14.1613381	99.552982	99.5529807	99.5529557	99.5695708
$p_{1,3}$	-3015.5257	-705.07937	-3015.5258	-3015.5257	-3015.5255	-3015.524
$p_{1,2}$	-31135.091	-3783.072	-31135.092	-31135.091	-31135.089	-31135.651
$p_{1,1}$	469207.016	86557.5176	469207.033	469207.019	469206.994	469204.673

$p_{1,0}$	-44152.655	-6375.9745	-44152.655	-44152.654	-44152.657	-44147.375
$p_{2,1}$	-2068410.4	15826.7839	-2068410.5	-2068410.4	-2068410.2	-2068176.3
$p_{2,0}$	37488220.6	-295844.63	37488221.8	37488220.9	37488217.3	37483858.9
$p_{3,3}$	266102.063	-659259758	266102.114	266102.076	266102.016	266088.379
$p_{3,2}$	-17413866	4.8621E+10	-17413869	-17413867	-17413863	-17412942
$p_{3,1}$	380664060	-1.195E+12	380664135	380664079	380663991	380643295
$p_{3,0}$	- 2.779E+09	9.794E+12	- 2.779E+09	- 2.779E+09	- 2.779E+09	- 2.778E+09
d_1	17.2006	15.7525	17.2006	17.2006	17.2006	17.2121
d_2	19.7011	24.3518	19.7011	19.7011	19.7011	19.7076
"error _i "	yes	no	yes	yes	yes	no

The signal with different SNR values from 60 dB to 30 dB is chosen as the input. All other parameters in the identification process remain unchanged. Here, we choose to judge the validity of the identification results by the values of the discontinuity points d_1 and d_2 , and the results are shown in the table below. It can be observed that the identification error of the discontinuity point position remains constant between 50–60 dB. When the Signal-to-Noise Ratio (SNR) is below 50 dB, the error begins to increase. Finally, when the SNR reaches 30 dB, the identification algorithm fails to converge and becomes ineffective.

Table A2 The effect of the noise

S/N	True d_1	Identi. d_1	error	True d_2	Identi. d_2	error
60	17.2 mm	17.19 mm	0.06 %	19.7 mm	19.71 mm	0.05 %
55	17.2 mm	17.19 mm	0.06 %	19.7 mm	19.69 mm	0.05 %
50	17.2 mm	17.19 mm	0.06 %	19.7 mm	19.69 mm	0.05 %
45	17.2 mm	17.16 mm	0.23 %	19.7 mm	19.69 mm	0.05 %
40	17.2 mm	17.11 mm	0.52 %	19.7 mm	19.64 mm	0.30 %
35	17.2 mm	16.76 mm	2.56 %	19.7 mm	18.98 mm	3.65 %
30	17.2 mm	/	/	19.7 mm	/	/

Appendix B

Here, we additionally analyzed the cases where the penalty values are set to 1,000,000 and 10,000,000. The external excitation frequency is 9 Hz, and the external excitation amplitude is gradually increased from 0.005 mm to 4.5 mm; all other simulation settings are consistent with those in the paper, and the results are shown in the figure below. First, the external excitation amplitudes corresponding to the KCO structure's transition from intra-well oscillation around the first stable state, to intra-well oscillation around the second stable state, and finally to inter-well oscillation around the third stable state are identical (i.e., corresponding to the red dashed line, yellow dashed line, and green dashed line). The corresponding external excitation values are P1 (0.867 mm), P2 (0.913 mm), and P3 (0.958 mm), respectively. This clearly indicates that the penalty value has little impact on the bifurcation when the external excitation amplitude is small. As the external excitation increases, the evolution processes of the dynamic responses corresponding to the three penalty values are basically consistent, but differences exist at certain specific points. For example, at point A4 (3.092 mm), under a penalty value of 10,000, the KCO exhibits intra-well oscillation near the third stable state; however, in the other two cases, the KCO exhibits inter-well oscillation around the three stable points. This shows that the penalty value has a certain impact when the external excitation amplitude is large.

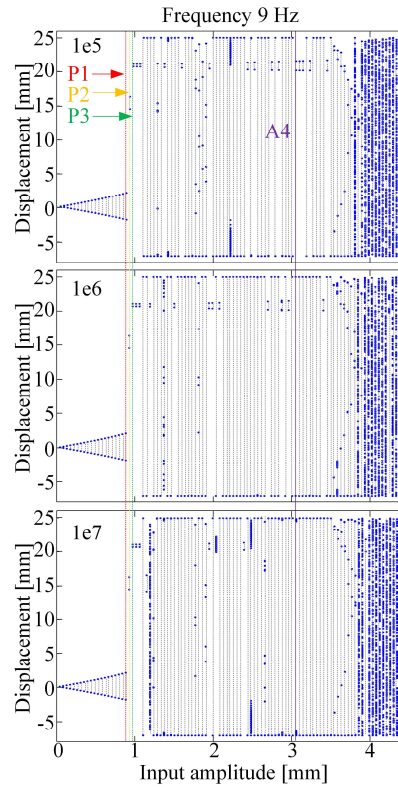


Fig. A1 Bifurcation diagrams of the KCO structure of three different penalty values: 100000, 1000000, and 10000000.

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